CHAPTER-8

REDOX REACTIONS

Oxidation	Reduction
1.Addition of oxygen	1.Removal of oxygen
2.Removal of hydrogen	2.Additionofhydrogen
3.Addition of an electronegative element	3.Removal of an electronegative element
4.Removal of an electropositive element	4.Addition of an electropositive element
5. Loss of electron	5. Gain of electron
6. Increasing oxidation number.	6, Decreasing oxidation number.

Oxidation number denotes the oxidation state of an element in a compound as curtained according to a set of rules formulated on the basis that electron in a covalent bond belongs entirely to more electro negative element .

Calculation of oxidation number

1. O.S. of all the element s in their elemental for m(in standard state) is taken as zero O.S. of element s in Cl_2,F_2, O_2 ,P_4, O_3,Fe(s),H_2, N_2, C(graphite) is zero.
2. Common O. S. of element s of group one (1st) is one. Common O. S. of elements of group two (2nd) is two.
3. For ions composed of only one atom, the oxidation number is equal to the charge on the ion.
4. The oxidation number of oxygen in most compound sis–2.While in peroxides (e.g., H_2O_2, Na_2O_2), each oxygen atom is assigned an oxidation number of -1, in super oxides (e.g.,KO_2 ,RbO_2) each oxygen atom is assigned an oxidation number of –(½).
5. In oxygen difluoride (OF_2) and dioxygen difluoride (O_2 F_2), the oxygen is assigned an oxidation number of +2 and +1, respectively.
6. The oxidation number of hydrogen is+1but in metal hydride its oxidation no. is–1.
7. In all its compounds, fluorine has an oxidation number of–1.
8. The algebraic sum of the oxidation number of all the atoms in a compound must be zero.
9. In poly atomic ion, the algebraic sum of all the oxidation numbers of atoms of the ion must equal the charge on the ion.

Stock notation: The oxidation number is expressed by putting a Roman numeral representing the oxidation number in parenthesis after the symbol of the metal in the molecular formula. Thus aurous chloride and auric chloride are written as Au (I) Cl and Au (III) Cl_3. Similarly, stannous chloride and stannic chloride are written as Sn (II) Cl_2 and Sn (IV)Cl_4.

Oxidation: An increase in the oxidation number

Reduction: A decrease in the oxidation number

Oxidizing agent: A reagent which can increase the oxidation number of an element in a given substance. These reagents are called as oxidants also.

Reducing agent: A reagent which lowers the oxidation number of an element in a given substance. These reagents are also called as reductants.

Redox reactions: reactions which involve change in oxidation number of the interacting species. Oxidation & reduction take place simultaneously.

Different types of reactions

1. Combination reaction – Two or more components combines to give a single product.

   $C\,(s) + O_2(g) \rightarrow CO2\,(g)$

2. Decomposition reaction – Breaking down of a compound into 2 or more components.

   $2H_2O\,(l) \xrightarrow{\Delta} 2H_2\,(g) + O_2\,(g)$

3. Displacement reaction – An ion/atom in a compound is replaced by an ion/atom of another element.

   $CuSO4\,(aq) + Zn(s) \xrightarrow{\Delta} Cu(s) + ZnSO4\,(aq)$

4. Disproportionation reaction – An element in one oxidation state is simultaneously oxidized reduced.

   $2H_2O_2\,(aq) \xrightarrow{\Delta} 2H_2O(l) + O_2(g)$

Balancing of redox reactions:

<u>Oxidation number Method</u>:

Write the net ionic equation for the reaction of potassium dichromate (VI), $K_2Cr_2O_7$ with sodium sulphite, Na_2SO_3, in an acid solution to give chromium (III) ion and the sulphate ion.

Step1: The skeletal ionic equation is:

$Cr_2O_7^{2-}(aq) + SO_3(aq) \rightarrow Cr(aq) + SO_4(aq)$

Step2: Assign oxidation number s for Cr and S

+6 -2 +4 -2 +3 +6 -2

$Cr_2O_7^{2-}(aq) + SO_3^{2-}(aq) \rightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$

Step3: Calculate the increase and decrease of oxidation number, and make the m equal:

+6 -2 +4 -2 +3 +6 -2

$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) \rightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq)$

Step4: balance the charge by adding H+ as the reaction occurs in the acidic medium,

$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq)\ 8H^+ \rightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq)$

Step5: balance the oxygen atom by adding water molecule.

$Cr_2O_7^{2-}(aq)+3SO_3^{2-}(aq)\ 8H^+ \rightarrow 2Cr^{3+}(aq)+3SO_4^{2-}(aq)+4H_2O(l)$

Half reaction Method

Balance the equations showing the oxidation of Fe^{2+} ions to Fe^{3+} ions by dichromate ions $(Cr_2O_7)^{2-}$ in acidic medium, where in,$Cr_2O_7^{2-}$ ions are reduced to Cr^{3+} ions .

Step1: Produce unbalanced equation for the reaction in ionic for m:

$Fe^{2+}(aq)+ Cr_2O_7^{2-}(aq) \rightarrow Fe^{3+}(aq)+Cr^{3+}(aq)$

Step2: Separate the equation into half reactions:

Oxidation half:

$$\overset{+2}{Fe^{2+}}(aq) \rightarrow \overset{+3}{Fe^{3+}}(aq)$$

Reduction half:

$$\overset{+6\ -2}{Cr_2O_7^{2-}}(aq) \rightarrow \overset{+3}{Cr^{3+}}(aq)$$

Step3: Balance the atom so the r than O and H in each half reaction individually.

$$Cr_2O_7^{2-}(aq) \rightarrow Cr^{3+}(aq)$$

Step4: For reactions occurring in acidic medium, add H_2O to balance O atoms and H_+ to balance H atoms.$Cr_2O_7^{2-}(aq) + 14H^+ \rightarrow Cr^{3+}(aq) + 7H_2O(l)$

Step5: Add electrons to one side of the half reaction to balance the charges. If need be, make the number of electrons equal in the two half reactions by multiplying one or both half reactions by appropriate coefficients.

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq)+ e-$

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

$6Fe^{2+}(aq) \rightarrow 6Fe^{3+}(aq) + 6e-$

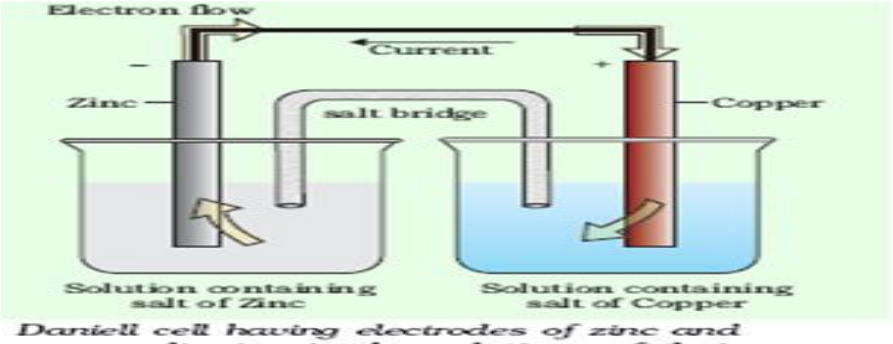
Step6: We add the two half reactions to achieve the overall reaction and cancel the electrons on each side. This gives the net ionic equation as:

$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$

A redox couple is defined as having together the oxidized and reduced forms of a substance taking part in an oxidation or reduction half reaction.

Represented as Zn^{2+}/Zn and Cu^{2+}/Cu.

- Electrochemical cells are the devices which are used to get electric current by using chemical reaction.



Daniell cell having electrodes of zinc and copper dipping in the solutions of their respective salts.

The potential associated with each electrode is known as electrode potential. If the concentration of each species taking part in the electrode reaction is unity (if any gas appears in the electrode reaction, it is confined to1atmospheric pressure) and further the reaction is carried out at 298K, the n the potential of each electrode is said to be the **Standard Electrode Potential**.

- SHE is used to measure electrode potential and its standard electrode potential is taken as 0.00V.

- **Electrochemical series-** A series of electrode potentials in either increasing or decreasing order is called electrochemical series. With the help of this series we can-

  (i) Find the emf of an electrochemical cell.

  (ii) Identify the reactivity series of elements in which they can displace others. Low electrode potential metal displaces the metal of high electrode potential and vice versa for non metals.

  (iii) Element having –ve electrode potential displaces H2 from dilute acids.

  (iv) Predict whether the redox reaction takes place or not

H.O.T.S information

- An element in its highest possible oxidation state in a compound behaves as an oxidizing agent.e.g. Mn in $KMnO_4$; N in HNO_3, Cl in $HClO_4$
- An element in its lowest possible oxidation state in a compound behaves as a reducing agent.e.g. S in Na_2SO_3; C in $H_2C_2O_4$; Sn in $SnCl_2$
- An element in its intermediate oxidation state in a compound behaves both as an oxidizing agent and reducing agent e.g. O in H_2O_2 ; N on HNO_2 ; S in SO_2
- An electronegative element in its lowest possible oxidation state in a compound or in free state behaves as a powerful reducing agent.e.g I^-,Br^- etc
- An electronegative element in its highest oxidation state in a compound behaves as a powerful oxidising agent.e.g Cl in $KClO_4$; Br in $KBrO_4$ etc

ONE MARK QUESTIONS

1. **Define oxidation and reduction in terms of oxidation number.**
   Ans. increase in oxidation number is oxidation and decrease in oxidation number is called reduction.
2. **What is meant by disproportionation ? Give one example.**
   Ans. In a disproportionation reaction an element simultaneously oxidized and reduced.
   $P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$
3. **What is O.N. of sulphur in H_2SO_4?**

   Ans: +6

4. **Identify the central atom in the following and predict their O.S. HNO_3**

   Ans. central atom:-N; O.S. +5

5. **Out of Zn and Cu which is more reactive?**

   Ans. Zn.

6. **What is galvanization?**

   Ans. Coating of a less reactive metal with Zinc. E.g.-coating of iron surface with Zn to prevent rusting of iron.

7. **How standard cell is potential calculated using standard electrode potential?**

   Ans: $E^0_{cell} = E^0_{cathode} - E^0_{anode}$

8. **What is O.S. of oxygen in H_2O_2?**

   Ans. -1.

9. **The formation of sodium chloride from gaseous sodium and gaseous chloride is a redox process. Justify.**

   Ans. Na atom gets oxidized and Cl is reduced.

10. **What will happen when chlorine is passed through an aqueous solution of potassium** bromide?
    Ans: The solution will acquire an orange colour due to vapours of bromine that are evolved.
    $2KBr + Cl_2 \rightarrow 2KCl + Br_2$

11. **Is the following reaction a redox reaction in nature? Justify your answer.**
    $Cr_2O_7^{2-} + H_2O \rightarrow 2CrO_4^{2-} + 2H^+$

    Ans . $\overset{+6\ -2}{Cr_2O_7^{2-}} + \overset{+1\ -2}{H_2O} \rightarrow \overset{+6\ -2}{2CrO_4^{2-}} + \overset{+1}{2H^+}$

    Since there is no change in oxidation no. of the elements involved in the reaction, it is not an example of redox reaction.

12. **Is $[Ag(NH_3)_2]^+ + 2H^+ \rightarrow Ag^+ + 2NH_4^+$ a disproportionation reaction?**
    Ans:

    $\overset{+1\ -3}{[Ag(NH_3)_2]^+} + 2H^+ \rightarrow \overset{+1}{Ag^+} + \overset{-3}{2NH_4^+}$

    Since there is no change in O.N. of the elements involved, it is not a disproportionation reaction.

13. **Why does fluorine not show disproportionation reaction?**
    Ans: Because its atom can only reduce by gaining an electron.

TWO MARKS QUEST IONS

1. **Write the balanced redox reaction.**

(I) $MnO_4^-(aq) + Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + Fe^{3+}(aq)$ **[acidic medium]**

(II) $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$ **[Acidic medium]**

Ans. (i) $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$

(ii) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

*** TRY TO BALANCE THE REDOX EQUATIONS IN ACIDIC AND BASIC MEDIUM BY BOTH THE METHODS.

2. **Identify the strongest & weakest reducing agent from the following metals:**

Zn, Cu, Na, Ag, Sn

Ans. Strongest reducing agent: Na, weakest reducing agent: Ag.

3. **Determine the oxidation no. of all the atoms in the following oxidants: $KMnO_4$, $K_2Cr_2O_7$ and $KClO_4$**

Ans:

In $KMnO_4$ K=+1, Mn = +7, O = -2

In $K_2Cr_2O_7$ K = +1, Cr = +6, O = -2

In $KClO_4$ K = +1, Cl = +7, O = -2

4. **Determine the oxidation no. of all the atoms in the following species: Na_2O_2 and OF_2.**

Ans. In Na_2O_2, Na = +1, O = -1

In OF_2, F = -1, O = +2

5. **Is it possible to store:**

(i) H_2SO_4 in Al container?

(ii) $CuSO_4$ solution in Zn vessel?

Ans. (i) Yes. (ii) No.

6. **Calculate the standard e.m.f. of the cell formed by the combination of**

$Zn/Zn^{2+}//Cu^{2+}/Cu$. Solution-: $E^0_{cell} = E^0_{cathode} - E^0_{anode}$

$= 0.34 - (-0.76) = 1.10V.$

7. **Identify the oxidizing and reducing agents in the following equations:**

i. $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$

ii. $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^{2+} \rightarrow 2Cr_2^{3+} + 6Fe^{3+} + 7H_2O$

Ans. (i) O.A. = MnO_4^-; R.A.= Fe^{2+}

(ii) O.A.= $Cr_2O_7^{2-}$; R.A.=Fe^{2+}

8. **Predict all the possible oxidation states of Cl in its compounds.**

Ans. 0, -1, +1, +3, +5, +7

9. Formulate possible compounds of 'Cl' in its O.S. is : 0,-1,+1,+3,+5,+7

Ans. Cl_2, HCl, HOCl, HOClO, $HOClO_2$, $HOClO_3$ respectively.

10. List three measures used to prevent rusting of iron.

Ans. (i) galvanization (coating iron by zinc)
(ii) greasing/oiling
(iii) Painting.

11. **H_2S acts only as reducing agent while SO_2 can act both as reducing agent and an** oxidizing agent. Discuss.

Ans.In H_2S, the oxidation state of S is -2.it can only increase its oxidation state but cannot decrease it. Therefore, it can act only as a reducing agent. In SO_2, the oxidation state of S is +4.It is in a position to undergo a decrease as well as an increase in the oxidation state. Thus SO_2 can act both as an oxidizing agent and a reducing agent

12. From the periodic table, select 2 non metals and 2 metals which can show disproportionation reactions.

Ans:Non metals:P_4,Cl_2

$P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$

$Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_20$

Metals:Cu^+,Ga^+

$2Cu^+ \rightarrow Cu + Cu^{2+}$

$3Ga \rightarrow Ga^{3+} + 2Ga$

13. Predict if the following reaction are feasible

a)Fe^{3+}(aq) and I^-(aq)

b)Br_2(aq) and Fe^{2+}(aq)

Ans. A particular reaction can be feasible if e.m.f of the cell based on the E° values being positive.

a) $I^- + Fe^{3+} \rightarrow Fe^{2+} + 1/2I_2$ E°cell=0.77-0.54=0.23V (feasible)

b)$Br^- + Fe^{3+} \rightarrow Fe^{2+} + 1/2Br_2$ E°cell=0.77-1.08=-0.31V(not feasible)

14. Predict the products of electrolysis of each of the following:

a) An aqueous solution of $AgNO_3$ using silver electrodes.

b) An aqueous solution of $AgNO_3$ using platinum electrodes

Ans:a) At cathode: $Ag(aq)^+ + e^- \rightarrow Ag$(deposited)

At anode: $Ag(s) \rightarrow Ag^+(aq) + e^-$

Thus the concentration of the solution will remain unaffected.

b) At cathode: $Ag\ (aq)^+ + e^- \rightarrow Ag$ (deposited)

At anode: OH^- ions with less discharge potential will be oxidized in preference to NO_3^- which will remain in solution.

$OH^-(aq) \rightarrow OH + e^-$;

$4OH \rightarrow 2H_2O(l) + O_2(g)$

Thus Ag is deposited on the cathode while O_2 is evolved at the anode. The solution will be acidic due to the presence of HNO_3

15. Write redox couples involved in the following reaction:

a) $Cu + Zn^{2+} \rightarrow Cu^{2+} + Zn$

b) $Br_2 + 2Cl^- \rightarrow Cl_2 + 2Br^-$

Ans: a) $Zn(s)|Zn^{2+}(aq);Cu^{2+}(aq)|Cu(s)$

b) $Cl_2(g)|2Cl^-(aq);2Br^-(aq)|Br_2(g)$

XI. Justify that the following redox reaction takes place or not-

$Ag(s) + CuSO_4(aq) \text{--------} \rightarrow Ag_2(SO4)(aq) + Cu(s)$

Ans- The above redox reaction does not take place because Ag has higher electrode potential than Cu, so it can not displace Cu from its solution.

XII, The reaction – $BaCl_2(aq) + Na_2SO_4(aq) \text{--------} \rightarrow 2NaCl(aq) + BaSO_4(s)$ is a redox reaction or not?

Ans- This reaction is not a redox reaction because in this reaction no change in oxidation number of any species occurs.

<u>THREE MARK QUESTIONS</u>

1. Writes Short notes on:

(a)Electrochemical series (b) Redox reactions (c) Oxidizing agents

Ans. (a) Electrochemical series: - arrangement of metals (non-metals also) in increasing order of their reducing power or vice versa.

(b) Reactions in which both oxidation and reduction take place simultaneously are REDOX REACTIONS.

(c) Oxidizing agents: chemical species which can oxidize the other one or can reduce itself.

2. Calculate O.S. of sulphur in the following oxoacids of 'S' :H_2SO_4, H_2SO_3, $H_2S_2O_8$ and $H_2S_2O_7$

Ans. +6, +4, +6 and +6 respectively.

(Calculate by considering x of 'S' and taking +1 of H, -2 0f 'O' and-1 of 'O' in peroxide bond.)

3. Explain role of salt bridge in Daniel cell.

Ans. (a) It completes the electric circuit in the cell.

(b) It maintains the electric neutrality in the cell.

*** EXPLAIN YOURSELF.

4. Account for the followings:

(i) Sulphur exhibits variable oxidation states.

Ans. Due to the presence of vacant 'd' orbital in 'S'

(ii) Fluorine exhibits only -1 O.S.

Ans. It is most electronegative element

(iii) Oxygen can't extend its valence from 2.

Ans. Small size/unavailability of vacant'd' orbital in O.S

5. Balance the equation $MnO4^- + I^- \rightarrow Mn^{2+} + I_2 + H_2O$ by ion electron method in acidic medium.

Ans. **Step-I** balancing of reduction half reaction by adding protons and electrons on LHS and more water molecules on RHS:

$8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Step-II Balancing of oxidation half reaction by adding electrons on RHS:

$2I^- \rightarrow I_2 + 2e^-$

Step-III To multiply the OHR by 5; RHR by 2 and to add OH & RH reactions to get overall redox reaction (cancellation of electrons of RH &OH reactions):

$[8H^+(aq) + MnO_4(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)]$ x 2

$[2I^- \rightarrow I_2 + 2e^-]$ x 5

$MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$

6. Complete and balance the following equations :

(i) $H^+ + Cr_2O_7^{2-} + Br^- \rightarrow 2Cr^{3+} + Br_2 +$ -----

(ii) $H_2O_2 + Cl^- \rightarrow OH^- + Cl_2$

(iii) $Zn + Cu^{2+} \rightarrow ?$

Ans. (i) $14H^+ + Cr_2O_7^{2-} + 6Br^- \rightarrow 2Cr^{3+} + 3Br_2 + 7H_2O$

(ii) $H_2O_2 + 2Cl^- \rightarrow 2OH^- + Cl_2$

(ii) $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

OTHER EXAMPLES OF BALANCING THE CHEMICAL REACTIONS CAN BE PRACTICED FROM TEXT BOOK.

7. Identify the oxidizing and reducing agents in the following equations:

(i) $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

(ii) $H_2 + Cl_2 \rightarrow 2HCl$

(iii) $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

Ans: (i) O.A.= H_2SO_4; R.A.=Fe

(ii) O.A.= Cl_2; R.A.=H_2

(iii) O.A.= MnO_2 ; R.A.=HCl

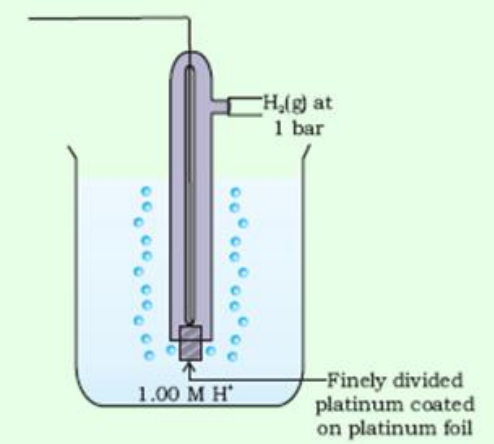
8. Arrange the following in increasing order of their reducing power :

Cu, Ag, Au, Zn, Fe, Al, Na, Mg, Pt (SHE), Hg, Ca, K

Ans: Au, Hg, Ag, Cu, Pt (SHE), Fe, Zn, Al, Mg, Na, Ca, K

9. What is SHE? What is its use?

Ans : Standard Hydrogen Electrode(SHE) has been selected to have zero standard potential at all temperatures. It consists of a platinum foil coated with platinum black(finely divided platinum) dipping partially into an aqueous solution in which the activity(approximate concentration 1M) of hydrogen ion is unity and hydrogen gas is bubbled through the solutionat1barpressure. The potential of the other half cell is measured by constructing a cell in which reference electrode is standard hydrogen electrode. The potential of the other half cell is equal to the potential of the cell.



10. Though alkaline potassium permangate and acidic potassium permanganate both are used as oxidants, yet in the manufacture of benzoic acid from toluene we use alcoholic potassium permanganate as an oxidant. Why? Write a balanced reaction.

Ans:

In acidic medium:

$5C_6H_5CH_3 + 6MnO_4^- + 18H^+ \rightarrow 5\ C_6H_5COOH + 6Mn^{2+} + 14H_2O$

In neutral (alcoholic) and alkaline medium:

$C_6H_5CH_3 + 2MnO_4^- \rightarrow 5\ C_6H_5COO^- + MnO_2 + H_2O + OH^-$

Neutral medium is preferred due to

- In the neutral medium neither acid nor base is to be added externally. This will definitely reduce the cost of the manufacturing process.
- Alcohol if used as solvent will help in the formation of a homogeneous mixture between toluene (non-polar) and $KMnO_4$ (ionic).

HOTS QUESTIONS

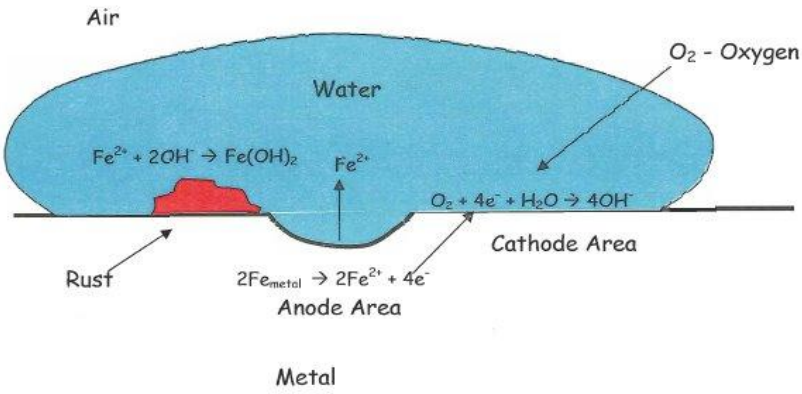
1. Is rusting of iron an electro chemical phenomenon? How? Explain.

Ans. Yes. Rusting of iron is an electrochemical phenomenon because this is possible due to formation of a small electrochemical cell over rough surface of iron and the following redox reaction takes place there in that cell-

Oxidation $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

$\frac{1}{2}\ O_2 + 2H_2O + 2Fe^{2+} \rightarrow Fe_2O_3 + 4H^+$



2. We spend crore of Rupees and even loose thousands of lives every year due to corrosion. How can we prevent it? Explain.

Ans: (i) By Galvanization: Coating of a less reactive metal with Zinc e.g.-coating of iron surface with Zn to prevent rusting of iron.

(ii) By greasing/oiling (to keep away the object from the contact of air & moisture.)

(iii)By painting (to keep away the object from the contact of air & moisture)

3. The compound AgF_2 is unstable compound. However if formed, it acts as a very strong oxidizing agent. Why?

Ans: In AgF_2 the oxidation state of Ag is +2 i.e. it exists as Ag^{2+} ion. This quite unstable since the normal stable of oxidation state of the metal is +1.this means that if the compound AgF_2 be formed it will undergo reduction by the gain of electrons. It is therefore a very strong oxidizing agent.

<u>Value based questions</u>:

1. A stain of rust is there on your cloth. You are worried how to remove this stain. Sita tells you to remove this stain using ripened guava.

a. Why?

b. What is the value you are having when doing this?

a. The rust is iron oxide. The oxalic acid in guava fruit dissolves iron Oxide.

b. Help your friends and neighbors when you know some simple home techniques instead of chemicals.

2. Ramesh went to a paper industry. The manager, paper industry insisted that H_2O_2 be used for bleaching instead of Chlorine in bleaching. Ramesh had learnt that Cl_2 is also a bleaching agent

a. Then why is H_2O_2 used instead of Cl_2

b. What value do you derive from this?

Ans:a) H_2O_2 after bleaching, the product formed is water. While when using chlorine the byproduct is HCl.

b) Seeing the products of the reaction we should select the reagents, so that there is minimum pollution

3. Arun bought a dry cell which was very old. He puts it in the torch. The torch did not glow. He found that the cell was dead. Its potential was zero.

a. Why did this happen?

b. What value did you derive from it?

Ans:a) As the dry cell (zinc and graphite) contains ions, over a period of time it gets consumed. Hence it gets completely used up even on long
Standing. b) Be thrifty and buy just the required amount of things.